

IRE 317

Control System Engineering

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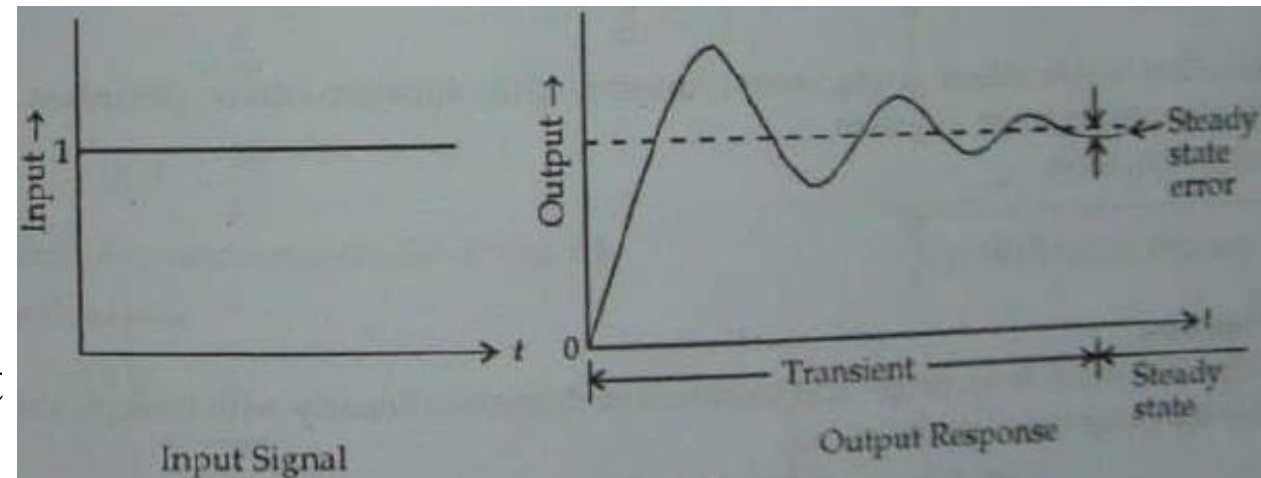
Time-Domain Analysis of Control Systems

Time-domain analysis studies how a control system's output varies with time when subjected to standard test inputs (step, ramp, impulse, etc.).

Response time is the time required for the system output to respond to a change in input and settle to its desired value.

The time response of a control system is divided into two parts (a) transient response, and (b) steady state response

Transient response is the part of the Response which goes to zero as the time increases and steady state response is the part of the total response after transient has died. If the steady state response of the output does not match with the input then the system has steady state error



Time-Domain Analysis of Control Systems

Mathematically, $c(t) = c_t(t) + c_{ss}(t)$

Where, $c_t(t)$ = transient response

$c_{ss}(t)$ = steady state response

Transient Response: It is the part of response which goes to zero as the time increases. Mathematically,

$$\lim_{t \rightarrow \infty} c_t(t) = 0$$

Steady State: It is the element of system response that is achieved when the time approaches to infinity. The response of the system after transient response is basically called steady state response. It is denoted by $C_{ss}(t)$.

Steady state error: If the steady state response of the output does not match with the input then the system has steady state error, denoted by e_{ss}

Time-Domain Analysis of Control Systems

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

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Test Signal for time Response

For the analysis of a control system the following inputs are used.

Step Function: Consider an independent voltage source in Series with a switch (s). When switch is open voltage is zero at the terminal.

Mathematically,

$$v(t) = 0 \quad ; \quad -\infty < t < 0$$

When the switch is closed at $t=0$

$$v(t) = K \quad ; \quad 0 < t < \infty$$

Combining above two equations

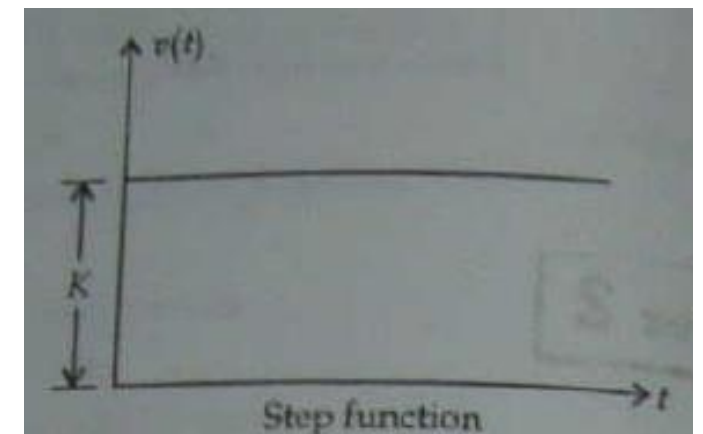
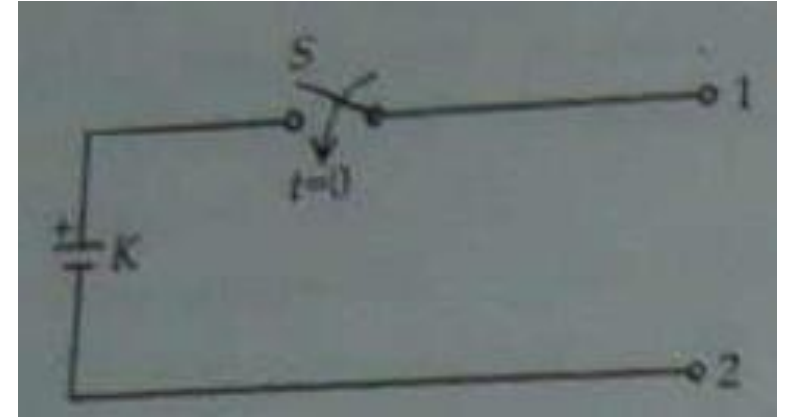
$$v(t) = 0 \quad ; \quad -\infty < t < 0$$

$$v(t) = K \quad ; \quad 0 < t < \infty$$

A unit step function is denoted by $u(t)$ and defined as

$$u(t) = 0 \quad ; \quad t \leq 0$$

$$u(t) = 1 \quad ; \quad 0 \leq t$$



Test Signal for time Response

Laplace Transform: Let, $f(t)$ be defined in the interval $0 \leq t \leq \infty$. L.T is obtained by multiplying $f(t)$ by e^{-st} and integrating between 0 to ∞ .

$$\mathcal{L}f(t) = \int_0^{\infty} u(t)e^{-st} dt = \int_0^{\infty} 1 \cdot e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{1}{s}$$

If input is $R(s)$ then $R(s) = \frac{1}{s}$

Test Signal for time Response

Ramp Function: It starts from origin and increases or decreases linearly with time.

Let, $r(t)$ be the ramp function.

$$r(t)=0 \quad t < 0$$

$$r(t)=kt \quad t > 0$$

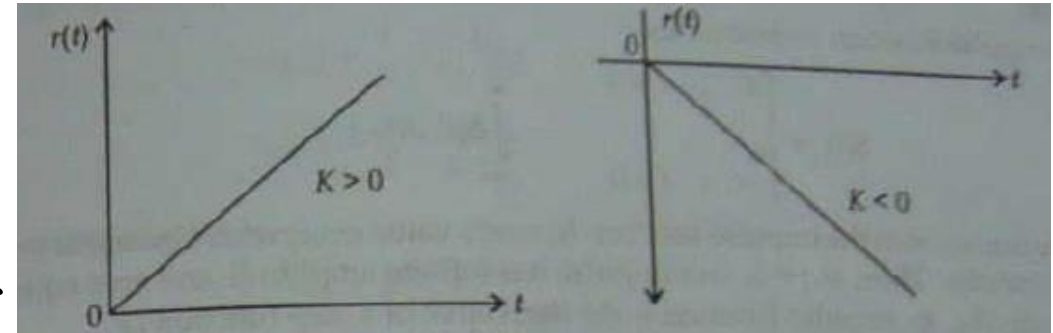
Where k is the slope of the line. For +ve value of k slope is upwards and -ve value of k slope is downwards.

Laplace Transform:

$$\mathcal{L}r(t) = \int_0^{\infty} r(t)e^{-st} dt = \int_0^{\infty} Kte^{-st} dt = \frac{K}{s^2}$$

$$\therefore R(s) = \frac{K}{s^2}$$

For unit ramp, $k=1$



Test Signal for time Response

Parabolic Function: The value of $r(t)$ is zero for $t < 0$ and is quadratic function of time for $t > 0$.

The parabolic function is defined as

LAPLACE TRANSFORM:

$$r(t) = 0 \quad t < 0$$

$$r(t) = \frac{Kt^2}{2} \quad t > 0$$

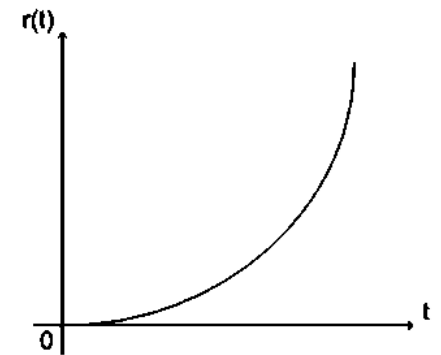
$$\mathcal{L}r(t) = \int_0^{\infty} r(t)e^{-st} dt = \int_0^{\infty} \frac{Kt^2}{2} e^{-st} dt = \frac{K}{s^3}$$

$$R(s) = \frac{K}{s^3}$$

For unit parabolic function $K=1$

$$r(t) = 0 \quad t < 0$$

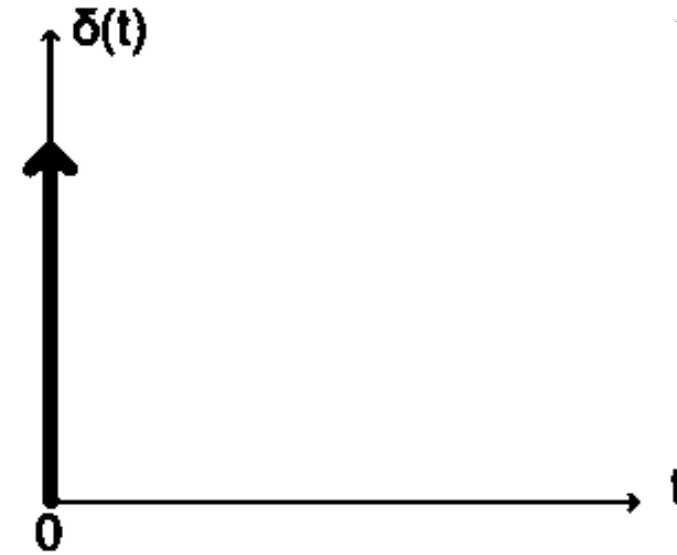
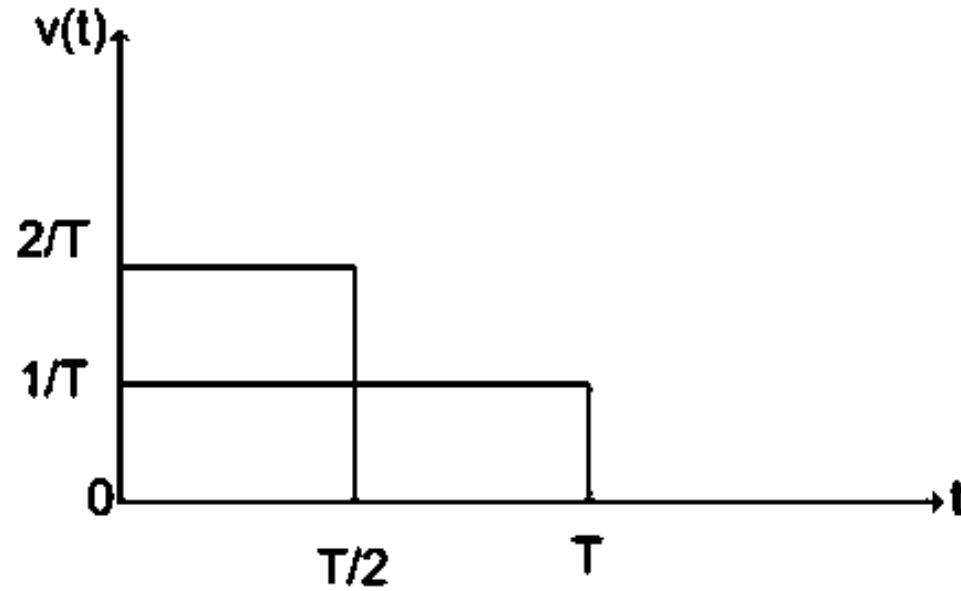
$$r(t) = \frac{t^2}{2} \quad t > 0$$



PARABOLIC FUNCTION

Test Signal for time Response

Impulse Response:



IMPULSE

The first pulse has a width T and height $1/T$, area of the pulse will be 1. If we halve the duration and double the amplitude we get second pulse. The area under the second pulse is also unity.

Test Signal for time Response

We can say as the duration of the pulse approaches to 0, the amplitude approaches to infinite but the area of the pulse remain unity.

The pulse for which the duration tends to zero and amplitude tends to infinity is called impulse. Impulse function also known as delta function. Mathematically

$$\begin{aligned}\delta(t) &= 0 && ; \quad t \neq 0 \\ &= \infty && ; \quad t = 0\end{aligned}$$

Thus the impulse function has zero value everywhere except at $t=0$, where the amplitude is infinite.

Test Signal for time Response

An impulse function is the derivative of a step function

$$\delta(t) = u(t)$$

$$\mathcal{L}\{\delta(t)\} = \mathcal{L}\left\{\frac{d}{dt}[u(t)]\right\} = s \cdot \frac{1}{s} = 1$$

INPUT	r(t)	SYMBOL	R(S)
UNIT STEP		U(t)	1/s
UNIT RAMP		r(t)	1/s ²
UNIT PARABOLIC		-	1/s ³
UNIT IMPULSE		$\delta(t)$	1

Laplace Transform Proof

Since

$$\delta(t) = \frac{du(t)}{dt},$$

using the Laplace differentiation property,

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0^-),$$

let $f(t) = u(t)$. Then

$$F(s) = \frac{1}{s}$$

$$u(0^-) = 0$$

Therefore,

$$\mathcal{L}\{\delta(t)\} = s\left(\frac{1}{s}\right) - 0 = 1.$$

Time response of a first order system with unit Step input

Consider a 1st order system with unity feed back.

For first order system

$$\frac{C(s)}{R(s)} = \frac{1}{sT + 1}$$

$$C(s) = \frac{1}{sT + 1} R(s)$$

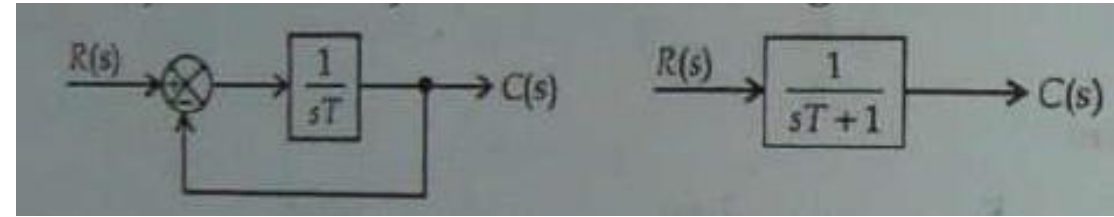
Input is unit step

$$R(s) = \frac{1}{s}$$

$$C(s) = \frac{1}{s(sT + 1)}$$

After partial fraction

$$C(s) = \frac{1}{s} - \frac{T}{1 + sT}$$



Time response of a first order system with unit Step input

Take inverse Laplace

$$c(t) = 1 - e^{-t/T}$$

When $t=T$ $c(t) = 1 - e^{-T/T} = 1 - e^{-1} = 0.632$

Where 'T' is known as 'time constant' and defined as the time required for the signal to attain 63.2% of final or steady state value.

Time constant indicates how fast the system reaches the final value.

Smaller the time constant, faster is the system response.

Time response of a first order system with unit ramp input

We know that

$$\frac{C(s)}{R(s)} = \frac{1}{sT + 1}$$

$$C(s) = \frac{1}{sT + 1} R(s)$$

Input is unit Ramp

$$R(s) = \frac{1}{s^2}$$

$$C(s) = \frac{1}{s^2(1 + sT)}$$

After partial fraction

$$C(s) = \frac{1}{s^2} - \frac{T}{s} + T \frac{1}{s + \frac{1}{T}}$$

Time response of a first order system with unit ramp input

Take inverse Laplace, we get

$$c(t) = t - T + Te^{-t/T}$$

Error signal

$$e(t) = r(t) - c(t)$$

$$e(t) = t - [t - T + Te^{-t/T}]$$

$$e(t) = T(1 - e^{-t/T})$$

$$\text{Steady state error} = \lim_{t \rightarrow \infty} (T - Te^{-t/T}) = T \quad \lim_{t \rightarrow \infty} e^{-t/T} = 0 (\text{for } T > 0)$$

The steady state error is equal to 'T', where 'T' is the time constant of the system.

For smaller time constant steady state error will be small and speed of the response will increase.

Time response of a first order system with unit impulse input

Input is unit impulse function $R(s)=1$

$$C(s) = \frac{1}{sT + 1} R(s)$$

$$C(s) = \frac{1}{sT + 1} \cdot 1$$

$$C(s) = \frac{1}{T} \frac{1}{s + 1/T}$$

Inverse Laplace transform

$$c(t) = \frac{1}{T} e^{-t/T}$$

Comparison of all three responses

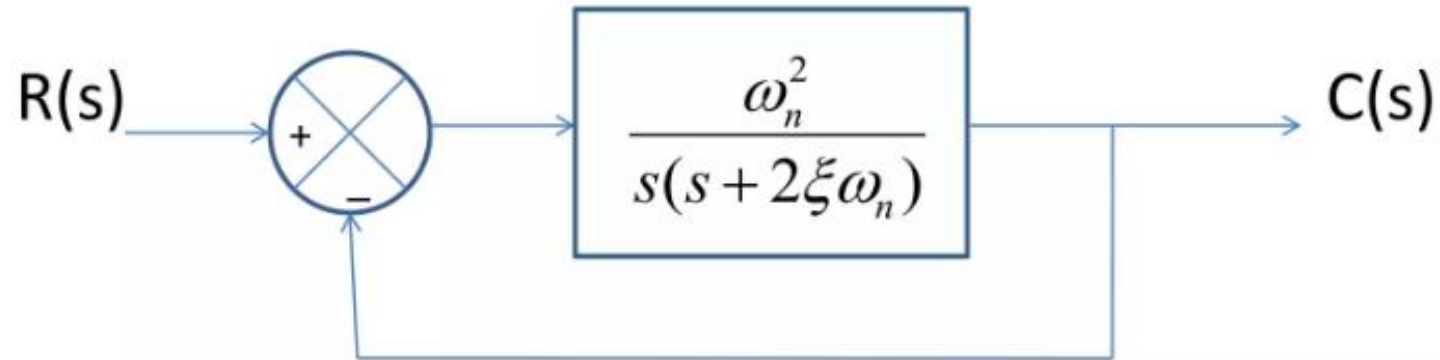
For unit Ramp Input $R(s) = \frac{1}{s^2}$ $c(t) = t - T + Te^{-t/T}$

For Unit Step Input $R(s) = \frac{1}{s}$ $c(t) = 1 - e^{-t/T}$

For Unit Impulse Input $R(s) = 1$ $c(t) = \frac{1}{T} e^{-t/T}$

It is clear that, unit step input is the derivative of unit ramp input and unit impulse input is the derivative of unit step input. This is the property of LTI system.

Time response of a 2nd order system with unit Step input



$$\frac{C(s)}{R(s)} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{-----}(A)$$

$$C(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} R(s)$$

For unit step input $R(s) = \frac{1}{s}$

Time response of a 2nd order system with unit Step input

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2} \text{-----} (1)$$

Replace $s^2 + 2\xi\omega_n s + \omega_n^2$ by $(s + \xi\omega_n)^2 + \omega_n^2(1 - \xi^2)$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s + \xi\omega_n)^2 + \omega_n^2(1 - \xi^2)} \text{-----} (2)$$

Break the equation by partial fraction and put $\omega_d^2 = \omega_n^2(1 - \xi^2)$

$$\frac{1}{s} \cdot \frac{\omega_n^2}{(s + \xi\omega_n)^2 + \omega_d^2} = \frac{A}{s} + \frac{B}{(s + \xi\omega_n)^2 + \omega_d^2} \text{-----} (3)$$

$$A = 1$$

Time response of a 2nd order system with unit Step input

Multiply equation (3) by $\left[(s + \xi\omega_n)^2 + \omega_d^2\right]$ and put

$$s = -\xi\omega_n - j\omega_d$$

$$B = \frac{\omega_n^2}{s}$$

$$B = \frac{\omega_n^2}{-\xi\omega_n - j\omega_d}$$

$$B = \frac{-\omega_n^2(\xi\omega_n - j\omega_d)}{(\xi\omega_n + j\omega_d)(\xi\omega_n - j\omega_d)}$$

$$-j\omega_d = s + \xi\omega_n$$

$$B = -(\xi\omega_n + s + \xi\omega_n) = -(s + 2\xi\omega_n)$$

Time response of a 2nd order system with unit Step input

Equation (1) can be written as

$$C(s) = \frac{1}{s} - \frac{s + \xi\omega_n + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} \qquad \frac{1}{s} \cdot \frac{\omega_n^2}{(s + \xi\omega_n)^2 + \omega_d^2} = \frac{A}{s} + \frac{B}{(s + \xi\omega_n)^2 + \omega_d^2}$$

$$C(s) = \frac{1}{s} - \left[\frac{s + \xi\omega_n}{(s + \xi\omega_n)^2 + \omega_d^2} + \frac{\xi\omega_n}{\omega_d} \cdot \frac{\omega_d}{(s + \xi\omega_n)^2 + \omega_d^2} \right] \quad \text{---(4)}$$

Laplace Inverse of equation (4)

$$c(t) = 1 - \left[e^{-\xi\omega_n t} \cdot \cos \omega_d t + \frac{\xi\omega_n}{\omega_d} e^{-\xi\omega_n t} \cdot \sin \omega_d t \right] \quad \text{---(5)}$$

Put $\omega_d = \omega_n \sqrt{1 - \xi^2}$

Time response of a 2nd order system with unit Step input

$$c(t) = 1 - e^{-\xi\omega_n t} \left[\cos\omega_d t + \frac{\xi}{\sqrt{1-\xi^2}} \sin\omega_d t \right]$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \left[\sqrt{1-\xi^2} \cos\omega_d t + \xi \sin\omega_d t \right]$$

Put $\sqrt{1-\xi^2} = \sin\phi$

$$\therefore \cos\phi = \xi$$

$$\tan\phi = \frac{\sqrt{1-\xi^2}}{\xi}$$

$$\phi = \tan^{-1} \left(\frac{\sqrt{1-\xi^2}}{\xi} \right)$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \phi)$$

Time response of a 2nd order system with unit Step input

Put the values of ω_d & ϕ

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\phi = \tan^{-1} \left(\frac{\sqrt{1 - \zeta^2}}{\zeta} \right)$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}} \sin \left[(\omega_n \sqrt{1 - \xi^2})t + \tan^{-1} \frac{\sqrt{1 - \xi^2}}{\xi} \right] \dots (6)$$

Error signal for the system

$$e(t) = r(t) - c(t)$$

$$e(t) = \frac{e^{-\xi\omega_n t}}{\sqrt{1 - \xi^2}} \sin \left[(\omega_n \sqrt{1 - \xi^2})t + \tan^{-1} \frac{\sqrt{1 - \xi^2}}{\xi} \right] \dots (7)$$

The steady state value of $c(t)$

$$\lim_{t \rightarrow \infty} c(t) = 1$$

Time response of a 2nd order system with unit Step input

Therefore, at steady state there is no error between input and output.

ω_n = natural frequency of oscillation or undamped natural frequency.

ω_d = damped frequency of oscillation.

$\xi\omega_n$ = damping factor or actual damping or damping coefficient.

For equation (A) two poles (for $0 \leq \xi \leq 1$) are

$$-\xi\omega_n + j\omega_n\sqrt{1-\xi^2}$$

$$-\xi\omega_n - j\omega_n\sqrt{1-\xi^2}$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Time response of a 2nd order system with unit Step input

Term	Definition	Effect
Pole	Denominator root	Stability, natural response
Zero	Numerator root	Shape/frequency of response

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

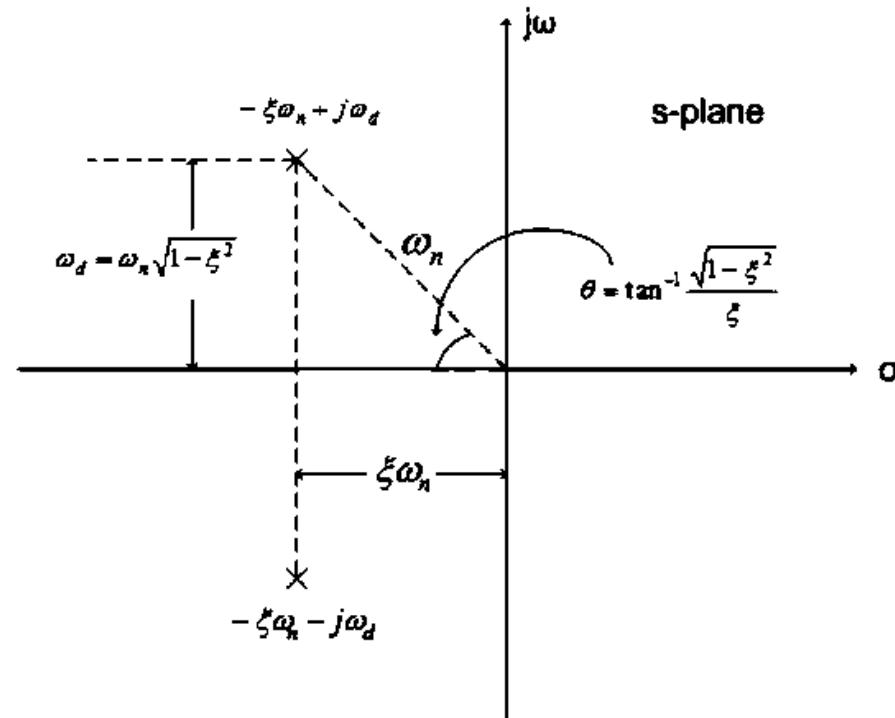
$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$s = -\zeta\omega_n \pm \omega_n \sqrt{\zeta^2 - 1}$$

Time response of a 2nd order system with unit Step input

Depending upon the value of ξ , there are 4 cases

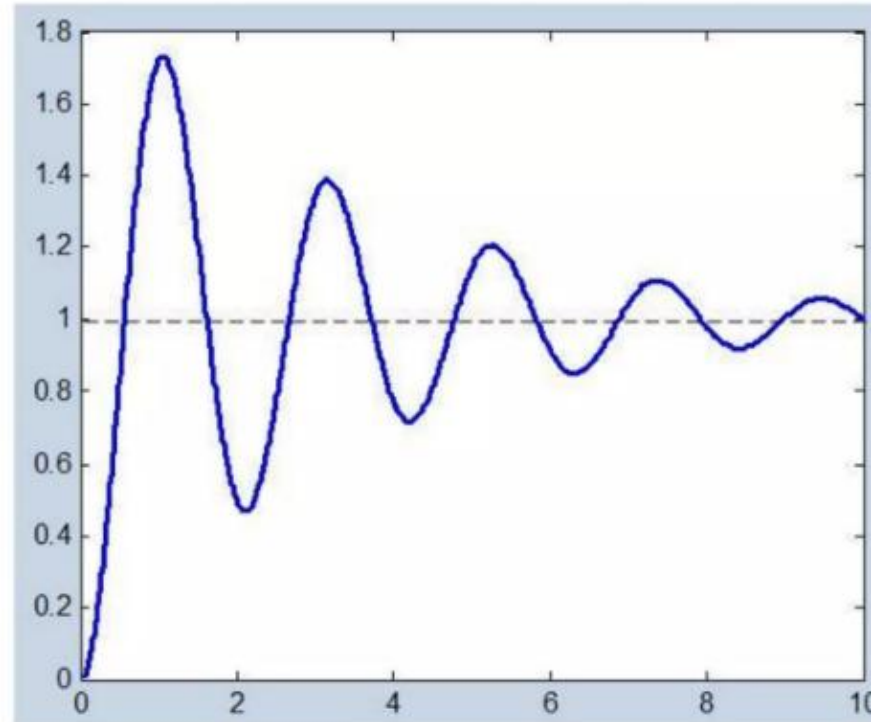
UNDERDAMPED ($0 \leq \xi \leq 1$): When the system has two complex conjugate poles.



Time response of a 2nd order system with unit Step input

From equation (6):

- Time constant is $1/\xi\omega_n$
- Response having damped oscillation with overshoot and under shoot. This response is known as under-damped response.

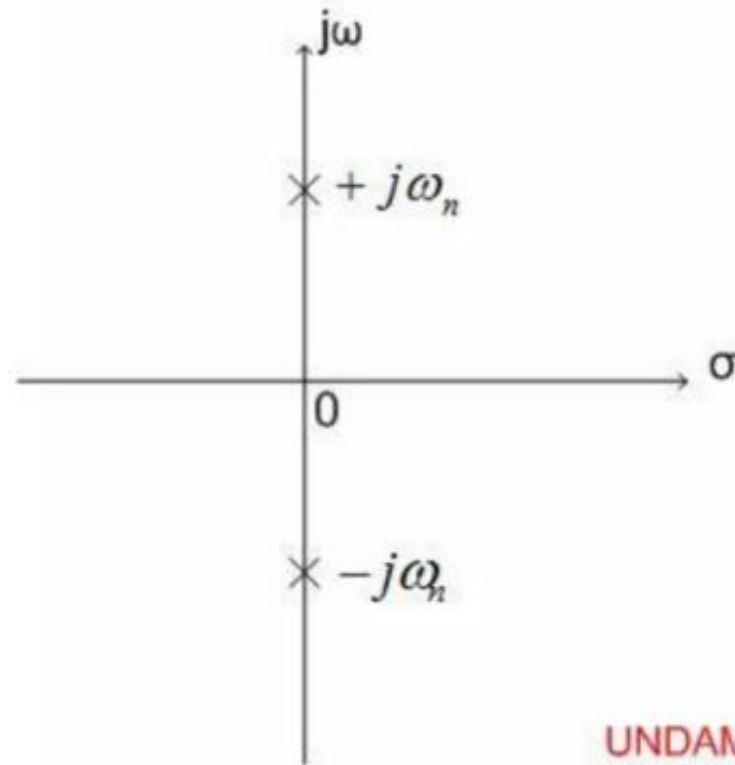


Time response of a 2nd order system with unit Step input

UNDAMPED ($\xi = 0$): when the system has two imaginary poles.

$$-\xi\omega_n + j\omega_n\sqrt{1-\xi^2}$$

$$-\xi\omega_n - j\omega_n\sqrt{1-\xi^2}$$



UNDAMPED CASE

Time response of a 2nd order system with unit Step input

From equation (6)

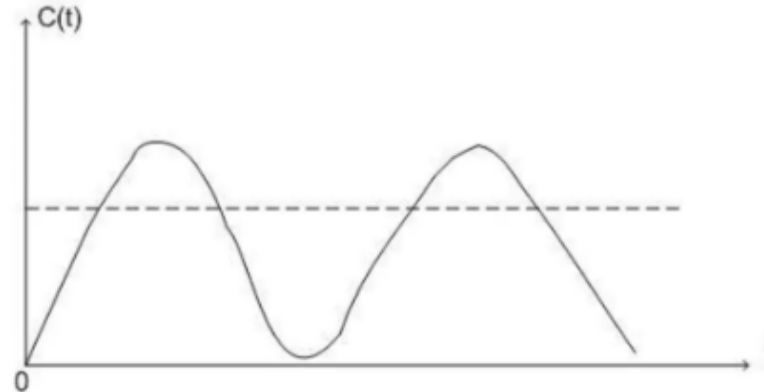
For $\xi = 0$

$$c(t) = 1 - \sin(\omega_n t + \pi / 2)$$

$$c(t) = 1 - \cos \omega_n t$$

Thus at ω_n the system will oscillate.

The damped frequency always less than the undamped frequency (ω_n) because of ξ . The response is shown in fig.



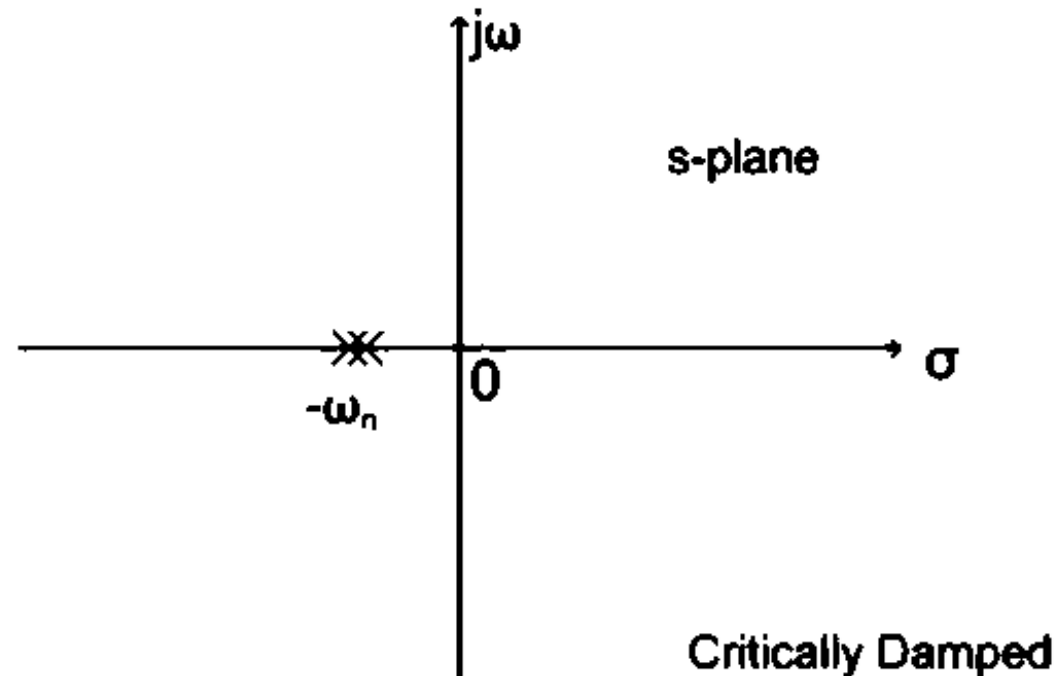
Sustain Oscillation

Time response of a 2nd order system with unit Step input

CRITICALLY DAMPED ($\xi = 1$): When the system has two real and equal poles. Location of poles for critically damped is shown in fig.

$$-\xi\omega_n + j\omega_n\sqrt{1-\xi^2}$$

$$-\xi\omega_n - j\omega_n\sqrt{1-\xi^2}$$



Time response of a 2nd order system with unit Step input

For $\xi = 1$

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$C(s) = \frac{\omega_n^2}{s(s + \omega_n)^2}$$

After partial
fraction

$$\frac{\omega_n^2}{s(s + \omega_n)^2} = \frac{1}{s} - \frac{1}{s + \omega_n} - \frac{\omega_n}{s(s + \omega_n)}$$

Take the inverse Laplace

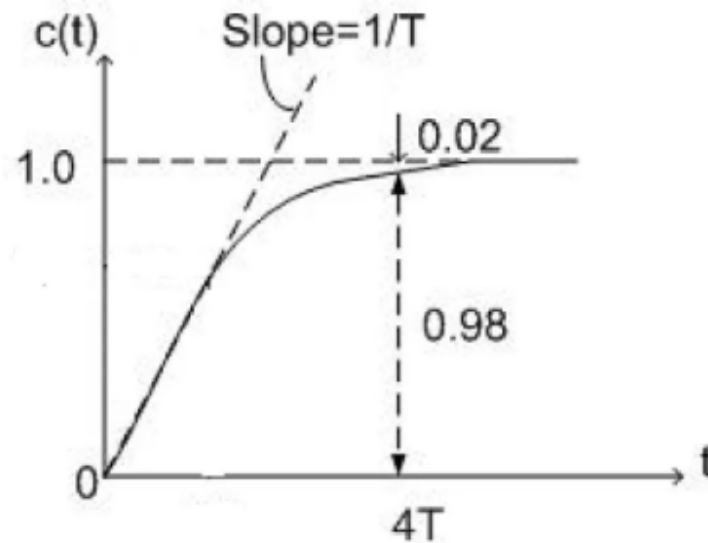
$$c(t) = 1 - e^{-\omega_n t} - t\omega_n e^{-\omega_n t}$$

$$c(t) = 1 - e^{-\omega_n t} (1 + \omega_n t) \dots (8)$$

Time response of a 2nd order system with unit Step input

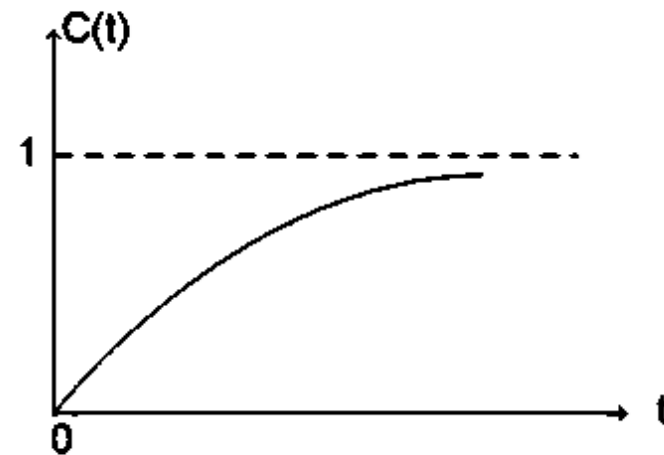
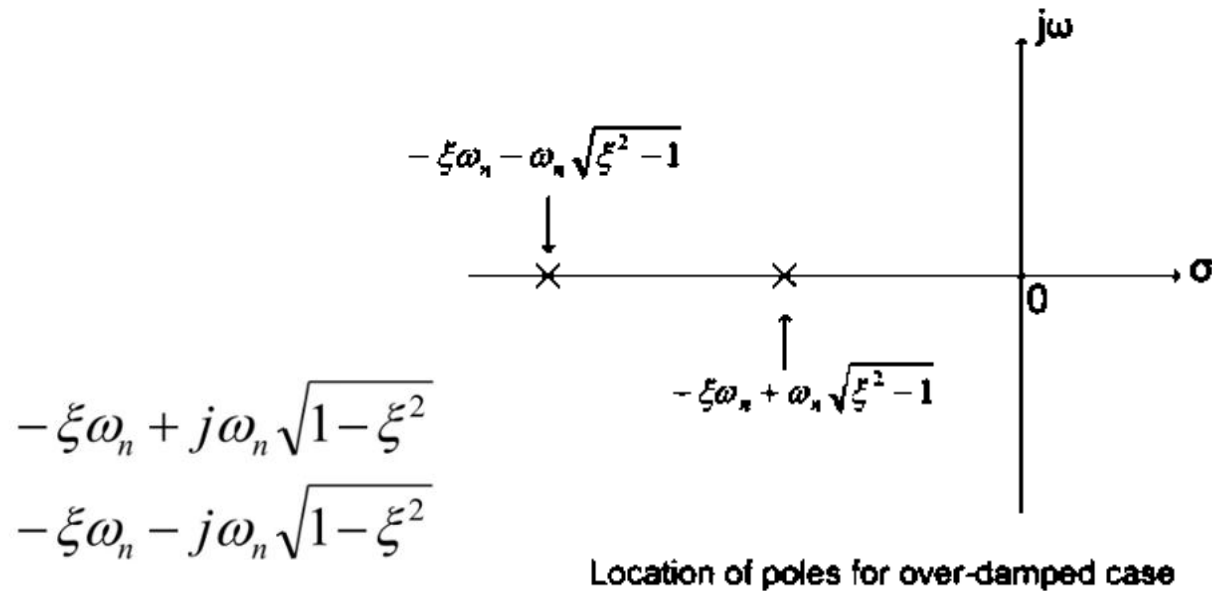
From equation (6) it is clear that $\xi\omega_n$ is the actual damping. For $\xi = 1$, actual damping $= \omega_n$. This actual damping is known as CRITICAL DAMPING.

The ratio of actual damping to the critical damping is known as damping ratio ξ . From equation (8) time constant $= 1 / \xi\omega_n$. Response is shown in fig.



Time response of a 2nd order system with unit Step input

Over damped ($\xi > 1$): When the system has two real and distinct poles.



Response of the system

Time response of a 2nd order system with unit Step input

From equation (2)

$$C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s + \xi\omega_n)^2 - \omega_n^2(\xi^2 - 1)} \text{ --- (9)}$$

$$\text{Put } \omega_d^2 = \omega_n^2(\xi^2 - 1)$$

$$\text{We get } C(s) = \frac{1}{s} \cdot \frac{\omega_n^2}{(s + \xi\omega_n)^2 - \omega_d^2} \text{ --- (10)}$$

Equation (10) can be written as

$$C(s) = \frac{\omega_n^2}{s(s + \xi\omega_n + \omega_d)(s + \xi\omega_n - \omega_d)} \text{ --- (11)}$$

Time response of a 2nd order system with unit Step input

After partial fraction of equation (11) we get

$$C(s) = \frac{1}{s} - \frac{1}{2\sqrt{\xi^2 - 1}(\xi + \sqrt{\xi^2 - 1})(s + \xi\omega_n + \omega_d)} - \frac{1}{2\sqrt{\xi^2 - 1}(\xi - \sqrt{\xi^2 - 1})(s + \xi\omega_n - \omega_d)} \quad \text{--- (12)}$$

Put the value of ω_d

$$C(s) = \frac{1}{s} - \frac{1}{2\sqrt{\xi^2 - 1}(\xi + \sqrt{\xi^2 - 1})(s + \xi\omega_n + \omega_n(\xi + \sqrt{\xi^2 - 1}))} - \frac{1}{2\sqrt{\xi^2 - 1}(\xi - \sqrt{\xi^2 - 1})(s + \xi\omega_n - \omega_n(\xi - \sqrt{\xi^2 - 1}))} \quad \text{--- (13)}$$

Time response of a 2nd order system with unit Step input

Inverse Laplace of equation (13)

$$c(t) = 1 + \frac{e^{-(\xi + \sqrt{\xi^2 - 1})\omega_n t}}{2\sqrt{\xi^2 - 1}(\xi + \sqrt{\xi^2 - 1})} - \frac{e^{-(\xi - \sqrt{\xi^2 - 1})\omega_n t}}{2\sqrt{\xi^2 - 1}(\xi - \sqrt{\xi^2 - 1})} \quad (14)$$

From equation (14) we get two time constants

$$T_1 = \frac{1}{(\xi + \sqrt{\xi^2 - 1})\omega_n}$$

$$T_2 = \frac{1}{(\xi - \sqrt{\xi^2 - 1})\omega_n}$$

Time response of a 2nd order system with unit Step input

From the eq. 14 it is clear that when ξ is greater than one there are two exponential terms, first term has time constant T_1 and second term has a time constant T_2 . $T_1 < T_2$. In other words we can say that first exponential term decaying much faster than the other exponential term. So for time response we neglect it, then

$$c(t) = 1 - \frac{e^{-(\xi - \sqrt{\xi^2 - 1})\omega_n t}}{2\sqrt{\xi^2 - 1}(\xi - \sqrt{\xi^2 - 1})} \text{ --- (15)}$$

$$T_2 = \frac{1}{(\xi - \sqrt{\xi^2 - 1})\omega_n} \text{ --- (16)}$$

Time response of a 2nd order system with unit Step input

Damping Ratio (ζ)	System Type	Step Response Behavior
($\zeta = 0$)	Undamped	Pure oscillation with constant amplitude
($0 < \zeta < 1$)	Underdamped	Oscillatory response with exponentially decaying amplitude
($\zeta = 1$)	Critically damped	Fastest rise to steady state without oscillation
($\zeta > 1$)	Overdamped	No oscillation, slow rise to steady state

Time domain specification of a 2nd order system

Delay time The delay time, t_d is defined as the time required for the step response to reach 50% of its final value.

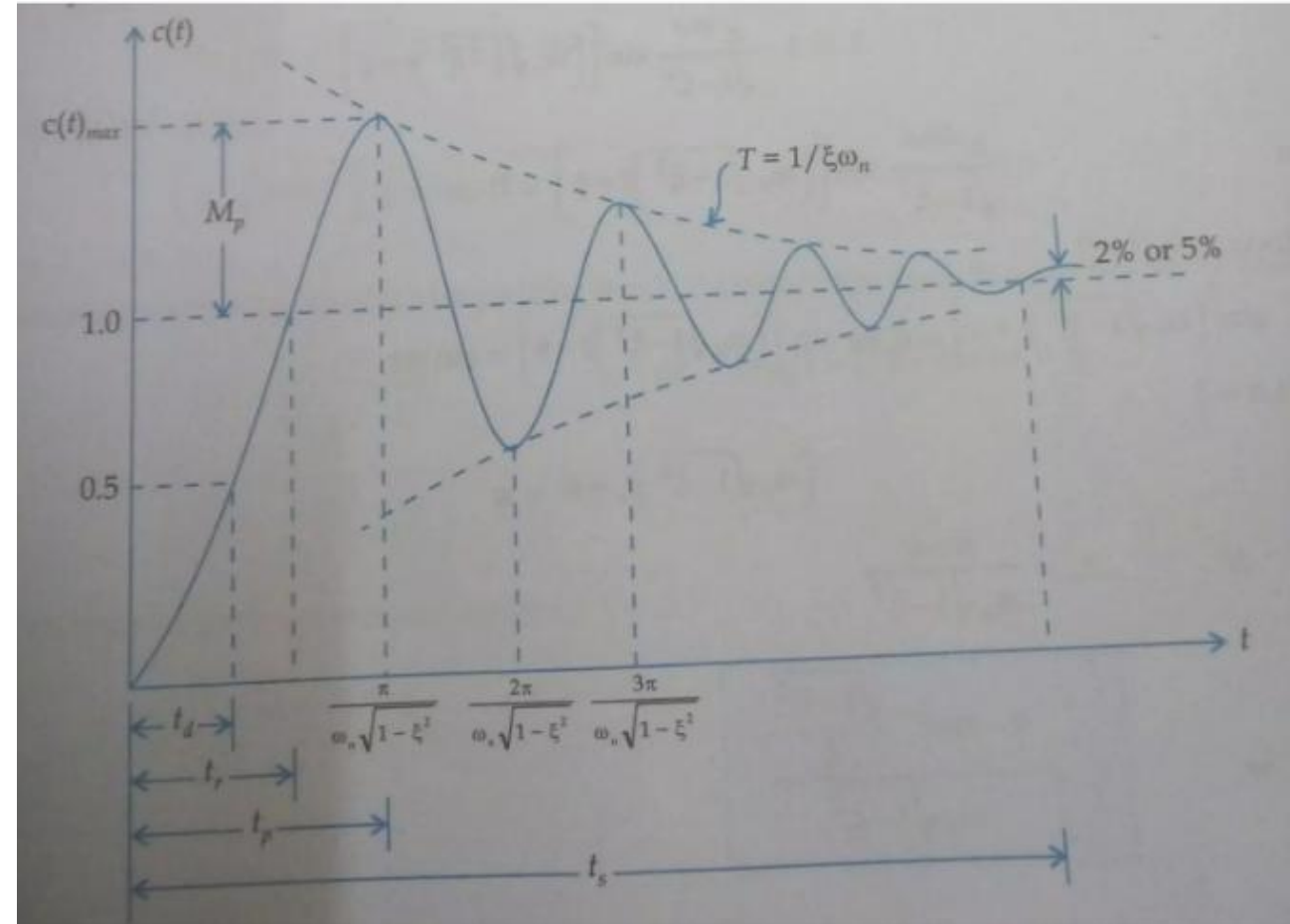
Rise time The rise time, t_r is defined as the time required for the step response to rise from 10 to 90 percent of its final value.

We know that:

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right]$$

Where,

$$\phi = \tan^{-1} \frac{\sqrt{1-\xi^2}}{\xi}$$



Time domain specification of a 2nd order system

Let, response reaches to 100% of the final value, then $c(t)=1$

$$1 = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right]$$

$$\frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right] = 0$$

Since, $e^{-\xi\omega_n t} \neq 0$

$$\sin((\omega_n \sqrt{1-\xi^2})t + \phi) = 0$$

Or, $\sin((\omega_n \sqrt{1-\xi^2})t + \phi) = \sin(n\pi)$

Put $n=1$

Time domain specification of a 2nd order system

$$(\omega_n \sqrt{1 - \xi^2}) t_r + \phi = \pi$$

Or,

$$t_r = \frac{\pi - \phi}{\omega_n \sqrt{1 - \xi^2}}$$

Or,

$$t_r = \frac{\pi - \tan^{-1} \frac{\sqrt{1 - \xi^2}}{\xi}}{\omega_n \sqrt{1 - \xi^2}}$$

Time domain specification of a 2nd order system

Peak Time (t_p): It is the time required for the response to reach the 1st peak of the time response.

Since

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right]$$

For maximum

$$\frac{dc(t)}{dt} = 0$$

$$\frac{dc(t)}{dt} = \frac{-e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \cos\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right] \omega_n \sqrt{1-\xi^2} +$$

$$\sin\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right] \frac{\xi\omega_n}{\sqrt{1-\xi^2}} e^{-\xi\omega_n t} = 0 \text{ --- (1)}$$

Time domain specification of a 2nd order system

Since, $e^{-\xi\omega_n t} \neq 0$

Equation can be written as

$$\cos\left[(\omega_n \sqrt{1-\xi^2})t + \phi\right] \sqrt{1-\xi^2} = \sin\left[(\omega_n \sqrt{1-\xi^2})t + \phi\right] \xi$$

Put $\sqrt{1-\xi^2} = \sin \phi$ and $\xi = \cos \phi$

Equation (2) becomes

$$\cos\left[(\omega_n \sqrt{1-\xi^2})t + \phi\right] \sin \phi = \sin\left[(\omega_n \sqrt{1-\xi^2})t + \phi\right] \cos \phi$$

$$\frac{\sin((\omega_n \sqrt{1-\xi^2})t + \phi)}{\cos((\omega_n \sqrt{1-\xi^2})t + \phi)} = \frac{\sin \phi}{\cos \phi}$$

Time domain specification of a 2nd order system

$$\tan((\omega_n \sqrt{1-\xi^2})t + \phi) = n\pi$$

The time to various peak
 $(\omega_n \sqrt{1-\xi^2})t_p = n\pi$

Where $n=1,2,3,\dots$

Peak time to first overshoot, put $n=1$

$$t_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}}$$

First minimum (undershoot) occurs at $n=2$

$$t_{\min} = \frac{2\pi}{\omega_n \sqrt{1-\xi^2}}$$

Time domain specification of a 2nd order system

Maximum Overshoot (Mp):

$$c(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin\left[\left(\omega_n \sqrt{1-\xi^2}\right)t + \phi\right]$$

Maximum overshoot occur at peak time, $t=t_p$

$$\text{Put, } t_p = \frac{\pi}{\omega_n \sqrt{1-\xi^2}} \text{ in above equation}$$

$$c(t) = 1 - \frac{e^{-\xi\omega_n \frac{\pi}{\omega_n \sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin\left[\cancel{\omega_n \sqrt{1-\xi^2}} \cdot \frac{\pi}{\cancel{\omega_n \sqrt{1-\xi^2}}} + \phi\right]$$

Time domain specification of a 2nd order system

$$c(t) = 1 - \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin(\pi + \phi)$$

$$c(t) = 1 + \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\sqrt{1-\xi^2}} \sin \phi \quad \because \sin(\pi + \phi) = -\sin \phi$$

Put, $\sin \phi = \sqrt{1-\xi^2}$

$$c(t) = 1 + \frac{e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}}{\cancel{\sqrt{1-\xi^2}}} \cancel{\sqrt{1-\xi^2}}$$

Time domain specification of a 2nd order system

$$c(t) = 1 + e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$$

$$M_p = c(t) - 1$$

$$M_p = 1 + e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} - 1$$

$$M_p = e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}}$$

$$M_p \% = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} * 100$$

Time domain specification of a 2nd order system

Settling time The settling time, t_s is defined as the time required for the step response to reach and stay within a specified percentage (5%) of its final value

Time consumed in exponential decay up to 98% of the input. The settling time for a second order system is approximately four times the time constant 'T'.

$$t_s = 4T = \frac{4}{\xi\omega_n}$$

STEADY STATE ERROR (e_{ss}): It is difference between actual output and desired output as time 't' tends to infinity.

$$e_{ss} = \lim_{t \rightarrow \infty} [r(t) - c(t)]$$

Time domain specification of a 2nd order system

Example 2.1. When a second order control system is subjected to a unit step input, the values of $\xi = 0.5$ and $\omega_n = 6$ rad/sec. Determine the rise time, peak time, settling time and peak overshoot.

Time domain specification of a 2nd order system

An open loop T/F of a system with unity feedback is given by

$$G(s) = \frac{10}{(s+2)(s+5)}$$

Determine the damping ratio, undamped natural frequency of oscillation. What is the percentage overshoot of the response to a unit step input.

SOLUTION: Given that

$$G(s) = \frac{10}{(s+2)(s+5)}$$

$$H(s) = 1$$

Characteristic equation $1 + G(s)H(s) = 0$

Time domain specification of a 2nd order system

An open loop T/F of a system with unity feedback is given by

$$G(s) = \frac{10}{(s+2)(s+5)}$$

Determine the damping ratio, undamped natural frequency of oscillation. What is the percentage overshoot of the response to a unit step input.

$$1 + \frac{10}{(s+2)(s+5)} = 0$$

$$s^2 + 7s + 20 = 0$$

Compare with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$ We get

$$\omega_n^2 = 20$$

$$2\xi\omega_n = 7$$

$$\therefore \omega_n = \sqrt{20} = 4.472 \text{ rad/sec}$$

$$2 * \xi * 4.472 = 7$$

$$\xi = 0.7826$$

$$M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} = e^{-\frac{\pi*0.7826}{\sqrt{1-(0.7826)^2}}} * 100 = 1.92\%$$

$$\omega_n 4.472 \text{ rad/sec}$$

$$\xi = 0.7826$$

$$M_p = 1.92\%$$

Time domain specification of a 2nd order system

A feedback system is given by the following T/F

$$G(s) = \frac{12}{s^2 + 4s + 16}$$

$$H(s) = Ks$$

The damping factor of the system is 0.8. determine the overshoot of the system and value of 'K'.

SOLUTION: We know that

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

$$\frac{C(s)}{R(s)} = \frac{16}{s^2 + (4 + 16K)s + 16}$$

$s^2 + (4 + 16K)s + 16 = 0$ is the characteristic eqn.

Time domain specification of a 2nd order system

A feedback system is given by the following T/F

$$G(s) = \frac{12}{s^2 + 4s + 16}$$

$$H(s) = Ks$$

The damping factor of the system is 0.8. determine the overshoot of the system and value of 'K'.

Compare with

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

$$\omega_n^2 = 16 \quad \therefore \omega_n = 4 \text{ rad/sec.}$$

$$2\xi\omega_n = 4 + 16K$$

$$2 * 0.8 * 4 = 4 + 16K \quad \therefore K = 0.15$$

$$M_p = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}}} * 100 = e^{-\frac{\pi 0.8}{\sqrt{1-(0.8)^2}}} * 100$$

$$M_p = 1.5\%$$

Time domain specification of a 2nd order system

The open loop T/F of a unity feedback control system is given by

$$G(s) = \frac{K}{s(1+sT)}$$

By what factor the amplifier gain 'K' should be multiplied so that the damping ratio is increased from 0.3 to 0.9.

SOLUTION:

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{K}{s(1+sT)}}{1+\frac{K}{s(1+sT)} \cdot 1}$$

$$\frac{C(s)}{R(s)} = \frac{K/T}{s^2 + \frac{s}{T} + \frac{K}{T}}$$

Characteristic Eq.

$$s^2 + \frac{s}{T} + \frac{K}{T} = 0$$

Time domain specification of a 2nd order system

The open loop T/F of a unity feedback control system is given by

$$G(s) = \frac{K}{s(1+sT)}$$

By what factor the amplifier gain 'K' should be multiplied so that the damping ratio is increased from 0.3 to 0.9.

Compare the characteristic eq. with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

We get

$$2\xi\omega_n = \frac{1}{T}$$
$$\omega_n^2 = \frac{K}{T} \quad \therefore \omega_n = \sqrt{\frac{K}{T}}$$

$$2\xi\sqrt{\frac{K}{T}} = \frac{1}{T}$$

Or,

$$\xi = \frac{1}{2\sqrt{KT}}$$

Given that:

$$\xi_1 = 0.3$$

$$\xi_2 = 0.9$$

$$\therefore \xi_1 = \frac{1}{2\sqrt{K_1 T}}$$

$$\xi_2 = \frac{1}{2\sqrt{K_2 T}}$$

Time domain specification of a 2nd order system

The open loop T/F of a unity feedback control system is given by

$$G(s) = \frac{K}{s(1+sT)}$$

By what factor the amplifier gain 'K' should be multiplied so that the damping ratio is increased from 0.3 to 0.9.

$$\therefore \frac{\xi_1}{\xi_2} = \sqrt{\frac{K_2}{K_1}}$$

$$\frac{K_2}{K_1} = \left(\frac{0.3}{0.9}\right)^2 = \frac{1}{9}$$

$$K_1 = 9K_2$$

Hence, the gain K_1 at which $\xi = 0.3$ Should be multiplied

By 1/9 to increase the damping ratio from 0.3 to 0.9

Time domain specification of a 2nd order system

Example 2.17. Consider the system shown in fig. 2.22. Determine the value of k such that the damping ratio ζ is 0.5. Then obtain the rise time (t_r), peak time (t_p), maximum overshoot M_p and settling time t_s in the unit step response.

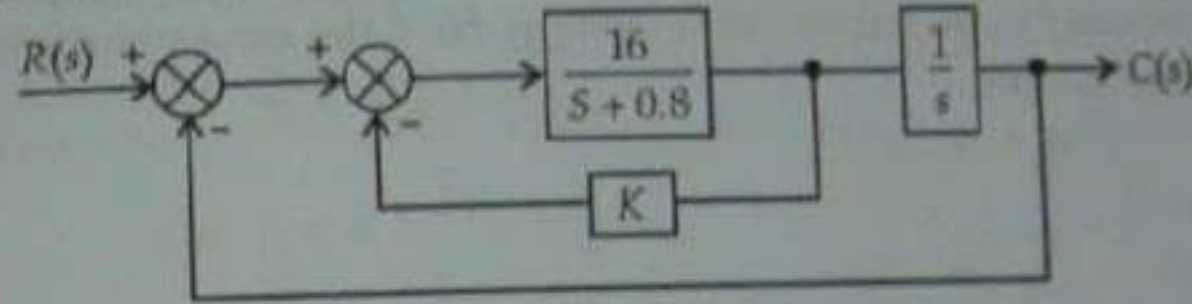


Fig. 2.22.

Time domain Analysis

What is Time-Domain Analysis?

Time-domain analysis studies how a system's output $y(t)$ responds to an input $u(t)$ over time.

Key aspects include:

1. Transient response – How the system reacts initially (rise time, overshoot, settling time).
2. Steady-state response – How the system behaves long-term.
3. Stability – Whether the system settles or diverges.

Importance in IoT

1.Real-time Monitoring & Response

- I. Many IoT devices (e.g., smart thermostats, health monitors) must react quickly and predictably to sensor data.
- II. Time-domain analysis ensures fast settling and minimal overshoot, preventing wrong actions or false alarms.

2.Signal Processing & Filtering

- I. IoT devices filter noisy signals from sensors.
- II. Understanding time-domain behavior ensures filters react properly without lag or instability.

3.Energy Efficiency

- I. Time-domain characteristics affect how long actuators (motors, valves) operate, impacting battery life.

Time domain Analysis

Importance in Robotics

1. Precise Motion Control

- I. Robots require exact position, velocity, or force tracking.
- II. Step and ramp responses help tune PID controllers to achieve desired speed and accuracy.

2. Stability & Safety

- I. Robots interact with humans and environments.
- II. Time-domain analysis ensures no oscillations or instability that can cause accidents.

3. Trajectory Planning & Dynamic Response

- I. For tasks like robotic arms or drones, the robot's output must follow desired trajectories.
- II. Time-domain metrics (rise time, settling time) help predict and optimize motion.